

Proposed Solution

Date	1 October 2026
Team ID	SWTID-2026-1235
Project Title	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

EduGenie uses Google Gemini generative AI to give learners instant, personalised explanations, practice questions and summaries through a simple web interface backed by a FastAPI service.

Section	Item	Details
Project Overview	Objective	Provide an AI learning assistant that makes studying faster and easier using Google Gemini.
	Scope	Web interface + FastAPI backend + Gemini integration for core learning modules; runs locally.
Problem Statement	Description	Learners struggle to find clear, personalised explanations and practice material quickly.
	Impact	Saves study time, improves understanding and builds confidence.
Proposed Solution	Approach	Prompt-engineered modules call the Gemini API and return structured, easy-to-read learning content.
	Key Features	- Ask Question module - Concept explanation module - Quiz / practice question module - Summary & study-plan module - Create diagram/Flowchart module - Live integration between web interface and backend API

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware – Computing	CPU/GPU	Processor: Intel i5 or equivalent (minimum)
Hardware – Memory	RAM	RAM: 4 GB
Hardware – Storage	Disk space	Storage: 128 GB SSD or 128 GB HDD
Hardware – Network	Internet connectivity	Internet Connectivity: High-speed internet (minimum 10 Mbps per system)
Hardware – Additional	Audio-visual setup	Additional: Audio-visual setup for interactive sessions (microphone, speakers, etc.)
Software – Frameworks	Python framework	FastAPI, Uvicorn
Software – Libraries	Additional libraries	google-generativeai, python-dotenv, pydantic, requests
Software – Browser	Web browser	Updated web browser (Google Chrome, Firefox, or Microsoft Edge)
Software – Environment	IDE	Visual Studio Code (or any preferred IDE)
Software – Version Control	Git	Git (latest version)
Data	Source, size, format	Generated on demand by Google Gemini API – no static dataset required